PURE Workshop 2026

Introduction to Data Visualization and Statistics

University of Kentucky

POLYPLOIDY INTEGRATION & INNOVATION INSTITUTE

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Why do we start here?

Your closest collaborator is you six months ago, and you do not reply to emails -Karl Broman

The most important tool is the mindset, when starting, that the end product will be reproducible - Keith Baggerly

Assume that everything that you are doing right now will need to be redone at some point in the future: be prepared

Five stages of reproducibility
by Claudia Solis-Lemus (UW Maddison)

Denial: I do not need to be reproducible. I have not kept track of code/scripts in years and I have been just fine.

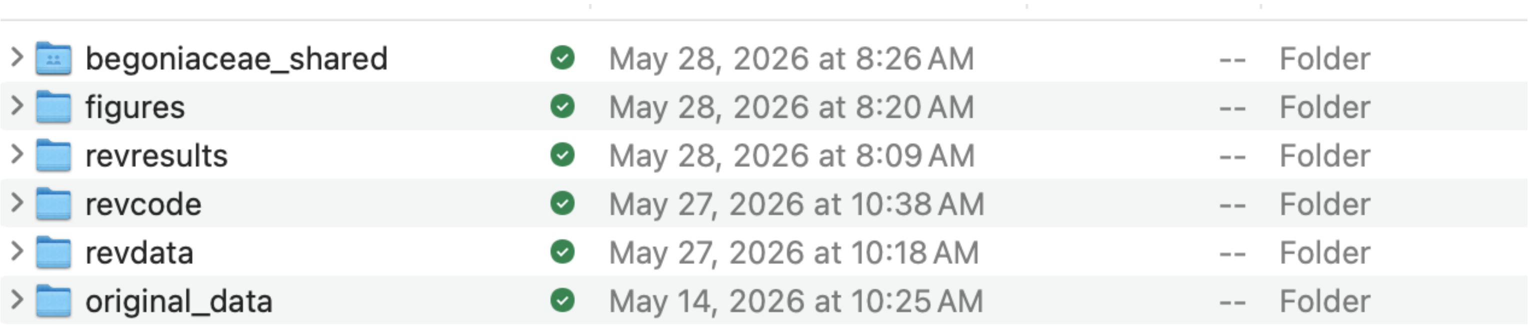
Anger: Why do I have to write these stupid notes in my data and code!? It takes twice the time to write notes and do the work. I could simply do the work!

Bargaining: Well, perhaps it is ok if I only keep notes in the very final script or the very final function. That makes sense. No one needs to know or would even care to read my other code.

Depression: I do not understand my notes. The comments that I made a year ago do not mean anything to me anymore.

Acceptance: I understand that being reproducible is a process. No one does this right the first time. No one does it right period.

Ways to keep organized your work



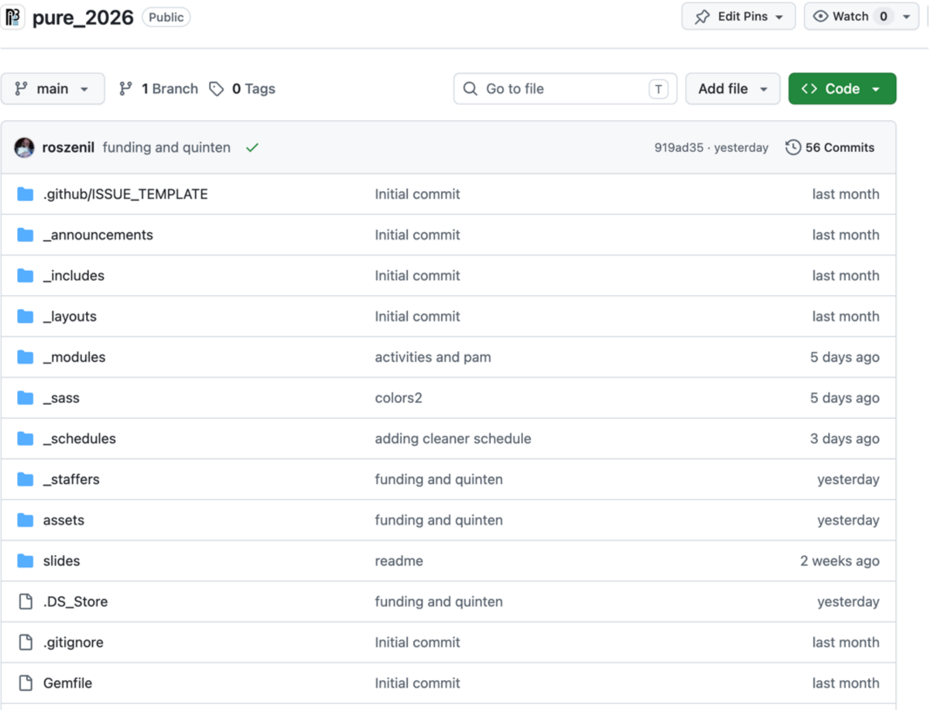
Cloud storage systems

- Backup in case your computer dies
- Automatic
- Private

Code version platforms

- Keep track of file changes
- Collaboration
- Big learning curve and aimed for big projects

GitHub



R Markdown and Rstudio

Reproducible notebook for statistics and figures

.Rmd files • Develop your code and ideas side-by-side in a single document. Run code as individual chunks or as an entire document.

Dynamic Documents • Knit together plots, tables, and results with narrative text. Render to a variety of formats like HTML, PDF, MS Word, or MS Powerpoint.

Reproducible Research • Upload, link to, or attach your report to share. Anyone can read or run your code to reproduce your work.

What we will be working today

01

Manipulating a dataset using a computer without modifying the original data

02

Making bar plots using a visualization tool called ggplot2

03

Making five-number summary statistics and plots for descriptive statistics

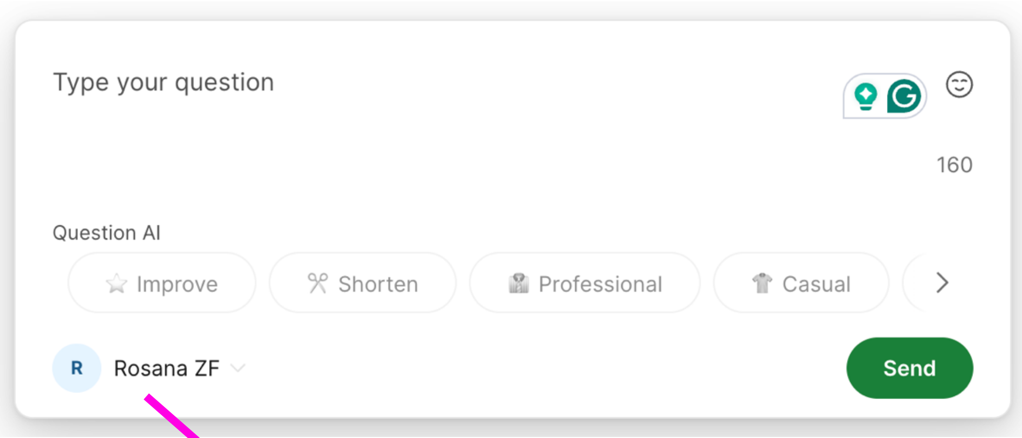
Background and dataset

Porturas et al. (2019) conducted literature surveys that integrated floral morphology, flowering phenology, and reproductive output with neopolyploidy.

We will compare **flower diameter** in neopolyploids and their parents.

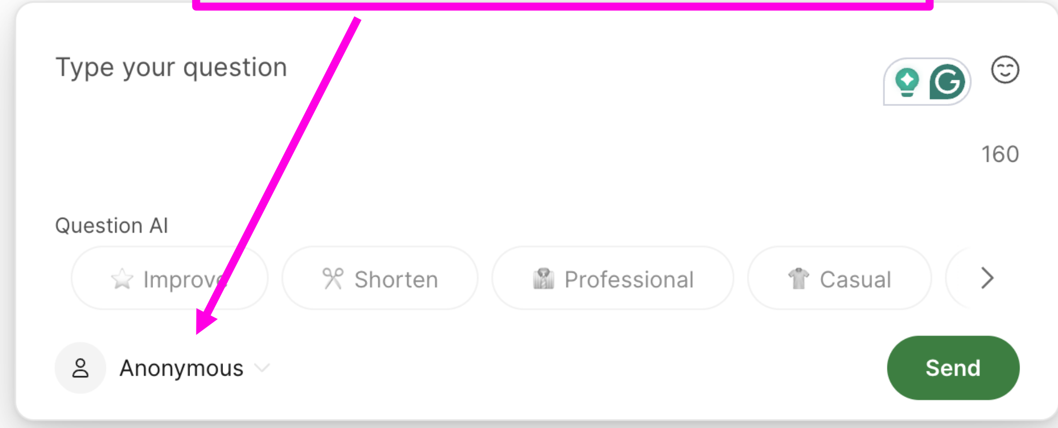
Testing **the gigas effect** = Polyploids have greater quantities of DNA and this causes larger cells and translates into larger tissues and organs

Ask questions!

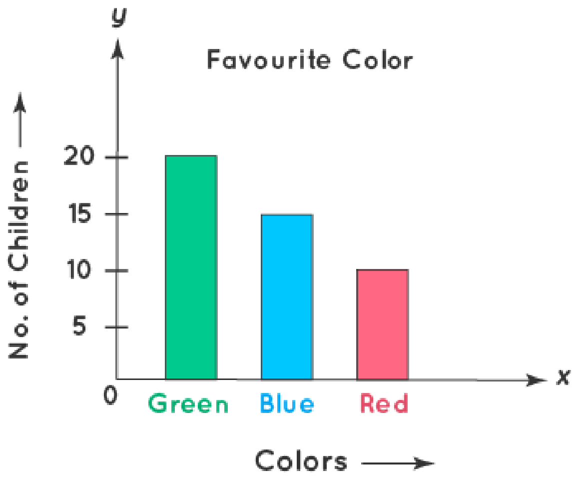


If you want us to go to your code in Posit.cloud please put your name (or if you want us to know you!)

Ask questions without judgement (just be polite)



Datasets are complicated



Vertical Bar Graph

Horizontal Bar Graph

A bar diagram is the graphical representation of precise data using rectangular bars with equal gaps between them.

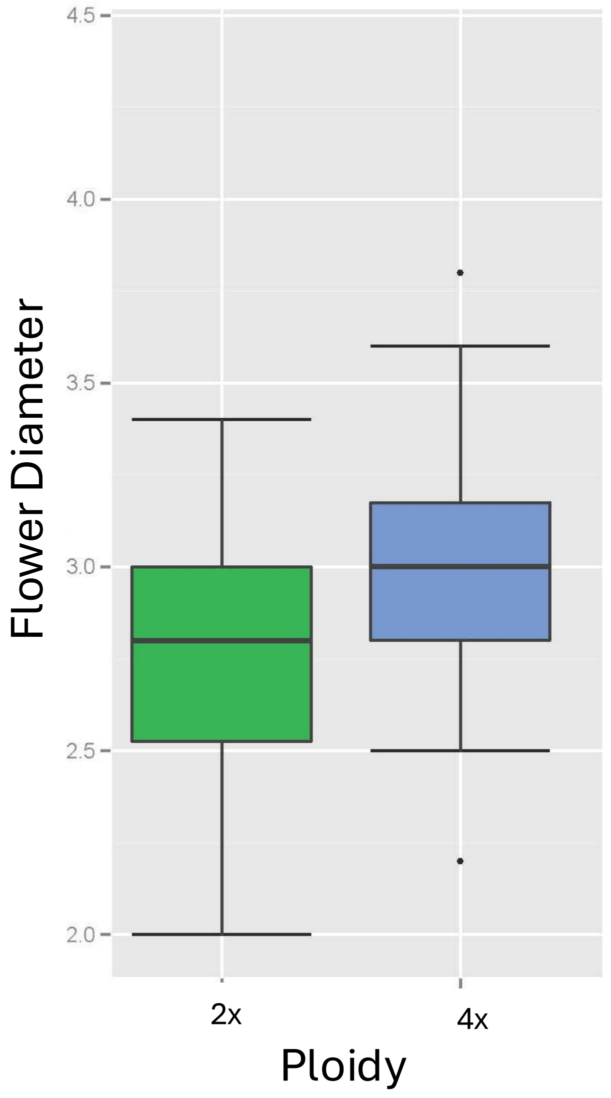
	Category	Total
This example	Colors	Children
Our example- RStudio	Traits	Studies

Choose one trait: Flower diameter

Are the flowers of neopolyploids bigger?

- Studies induced polyploidy using colchicine
- Measure diameter before treatment (2x) and after treatment (4x)

Flower diameter. Five-Point Summary



Box and Whisker plots

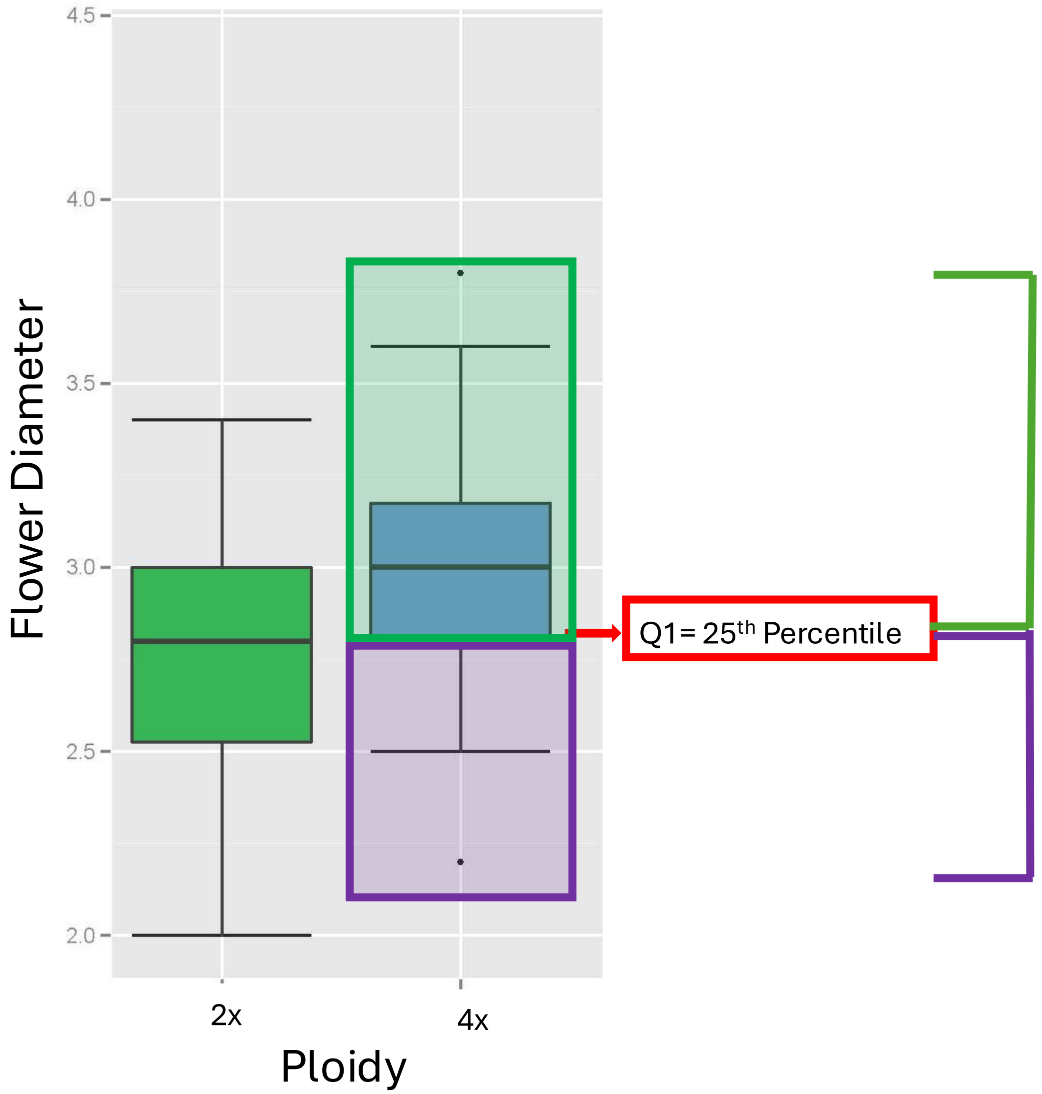
A box and whisker plot is defined as a graphical method of displaying variation in a set of data

Why do we need a visual?

An image is worth more than 100 words

We do not know if the distribution of flower diameters for neo-tetraploids is similar or has the same spread as the diploids

Flower diameter. Five-Point Summary

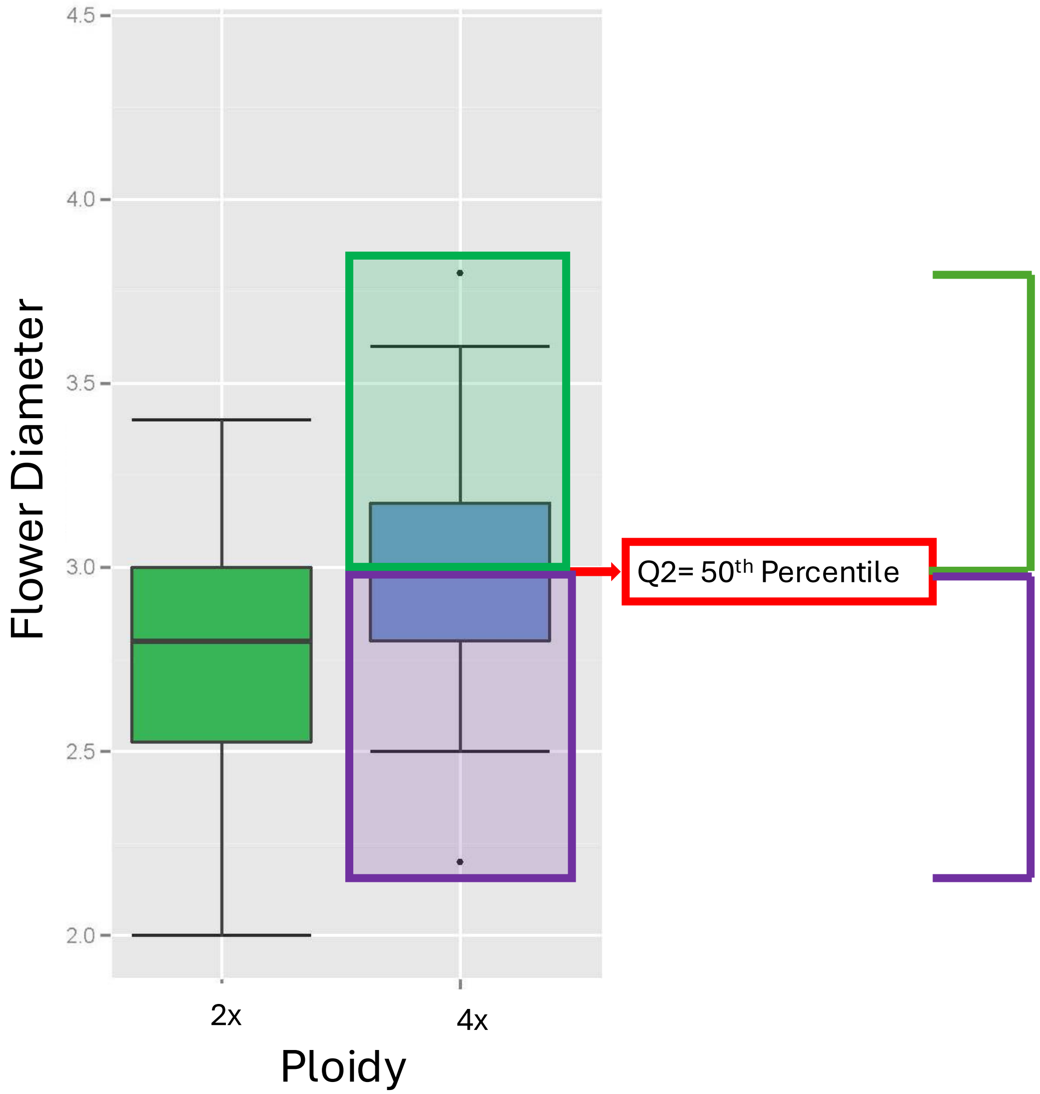


Q1 = 25th percentile

75% of all flower diameters are above this value

25% of all flower diameters are below this value

Flower diameter. Five-Point Summary

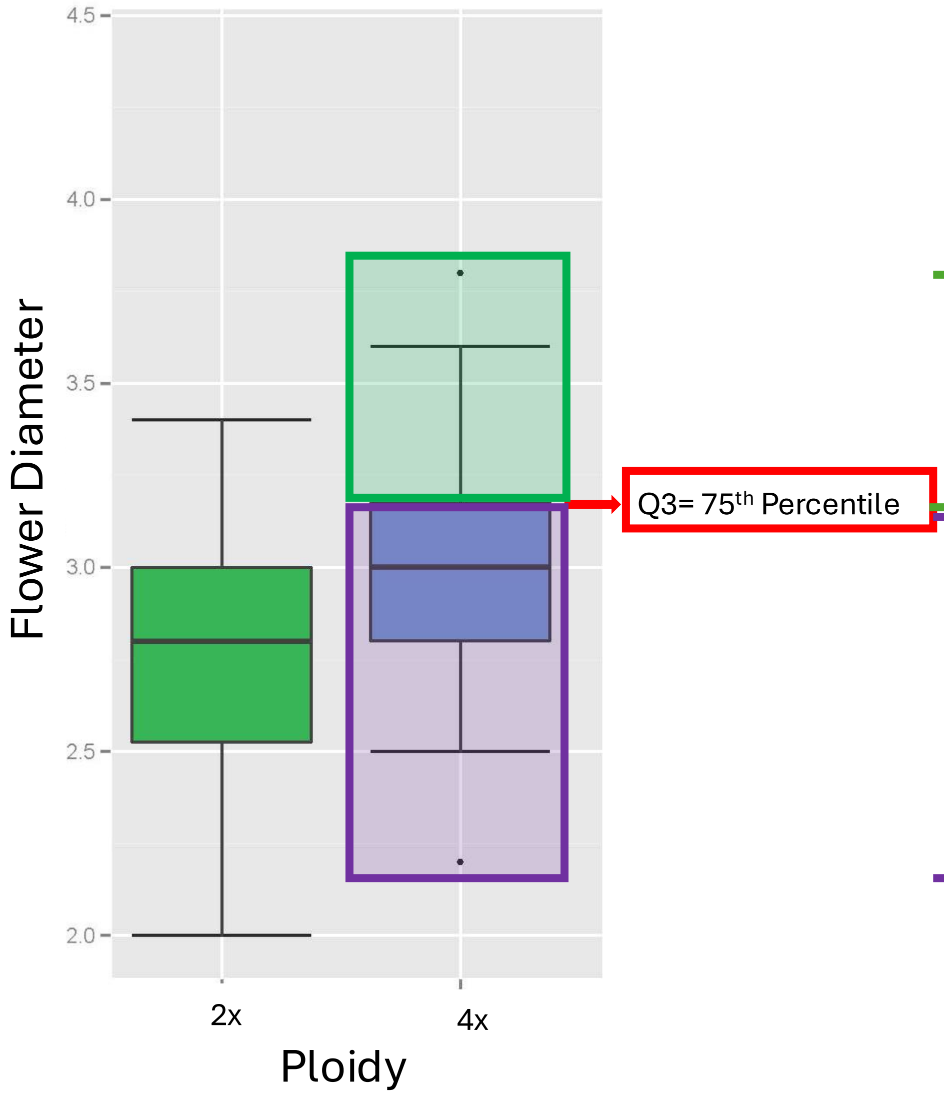


Q2 = 50th percentile= median

50% of all flower diameters are above this value

50% of all flower diameters are below this value

Flower diameter. Five-Point Summary

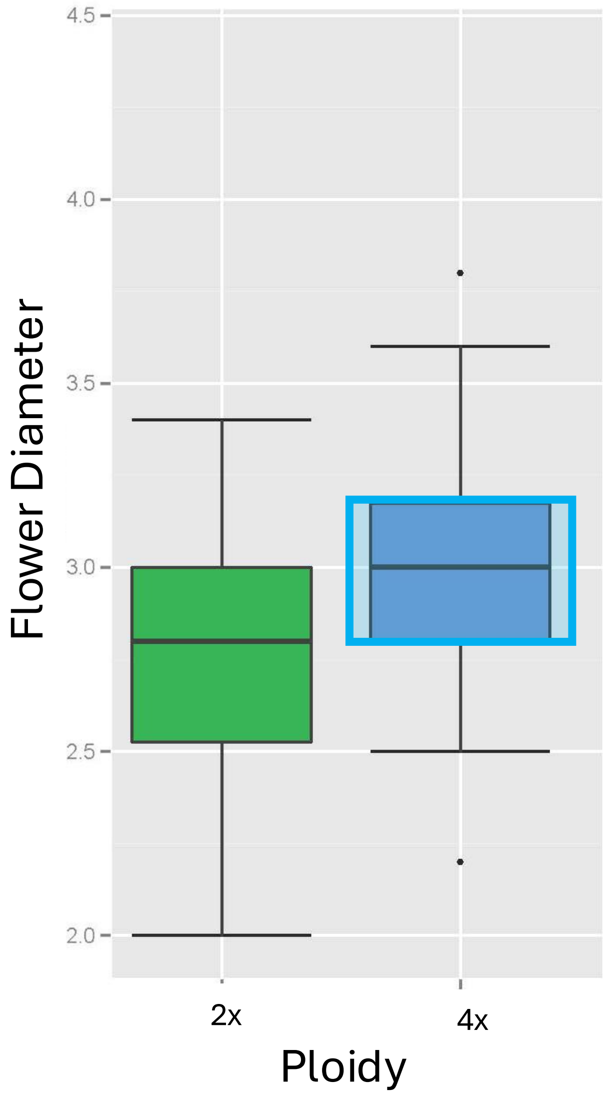


Q3= 75th percentile

25% of all flower diameters are above this value

75% of all flower diameters are below this value

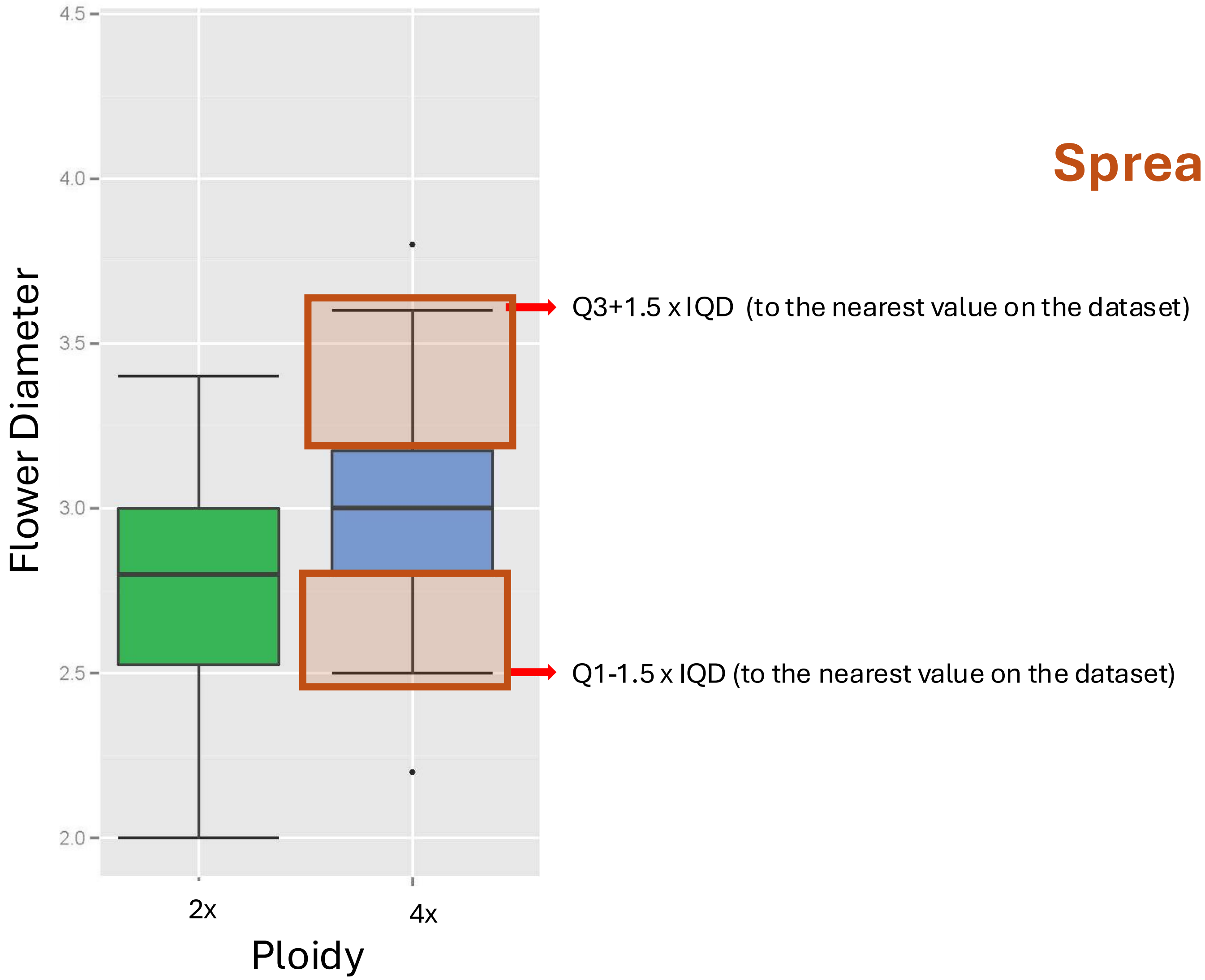
Flower diameter. Five-Point Summary



Interquartile Distance = IQR
=Q3-Q1

Interquartile Range
IQD
Are central values similar?

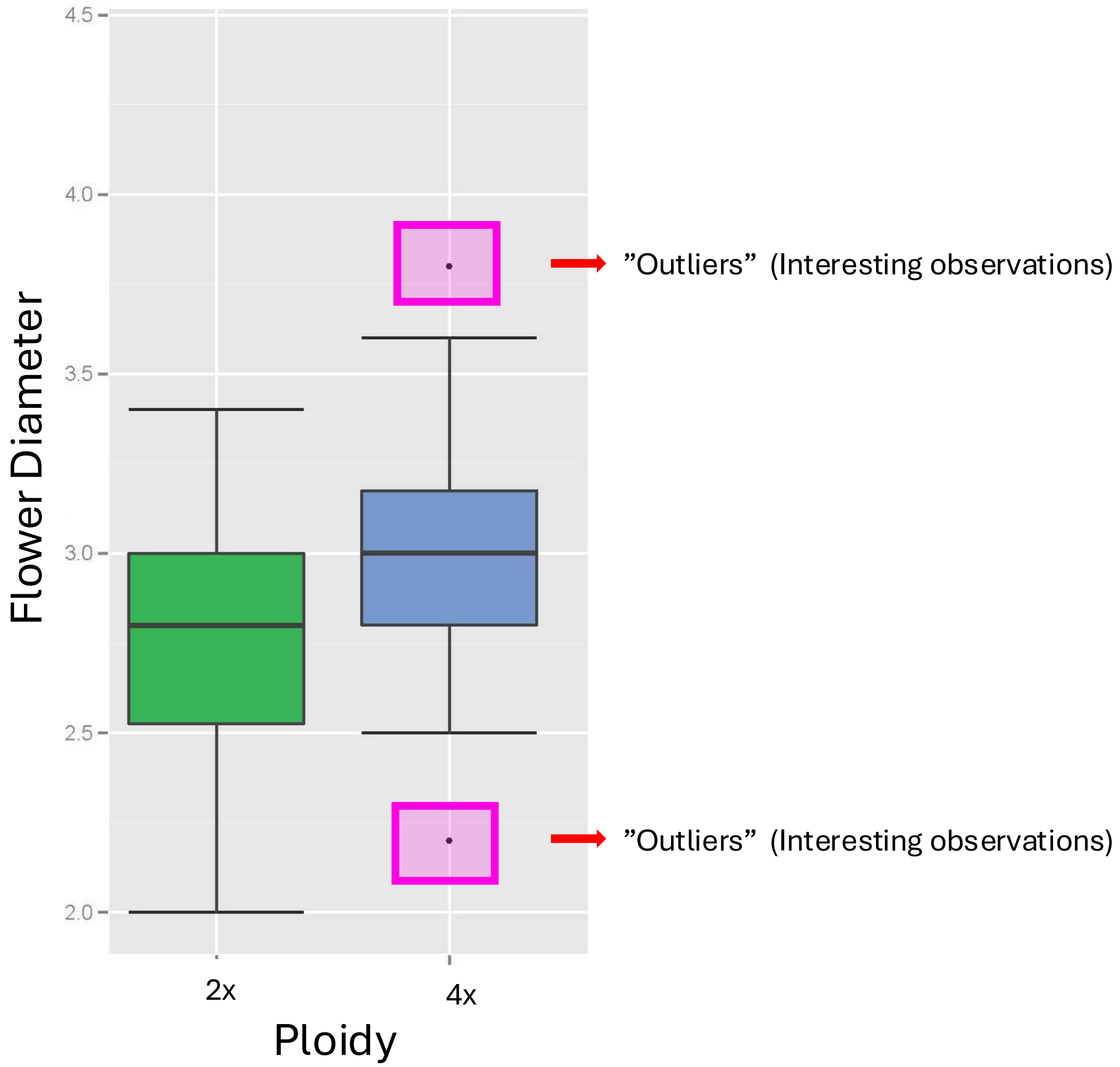
Flower diameter. Five-Point Summary



Whiskers

Spread of the distribution

Flower diameter. Five-Point Summary



Outliers
Interesting values worth exploring

(Maximum and Minimum sometimes)

Information and links

Workshop activities

Monday

06/29/2026:

INTRODUCTION — Slides

Doug Soltis and Pam Soltis. University of Florida

INTRODUCTION TO DATA VISUALIZATION — Setting up

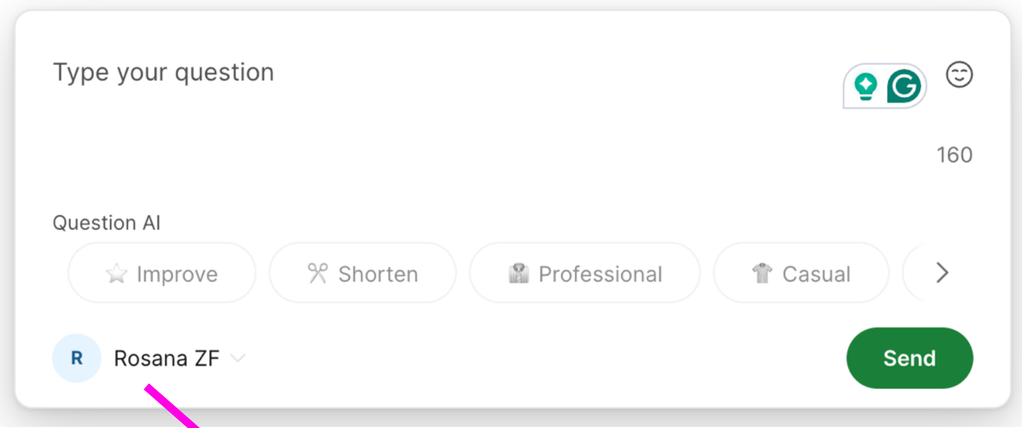
Activity — Link

Rosana Zenil-Ferguson and Pamela Rueda Cediel. University of Kentucky

INTRODUCTION TO PROTEOMICS — Link

Sixue Chen. University of Mississippi

Ask questions!



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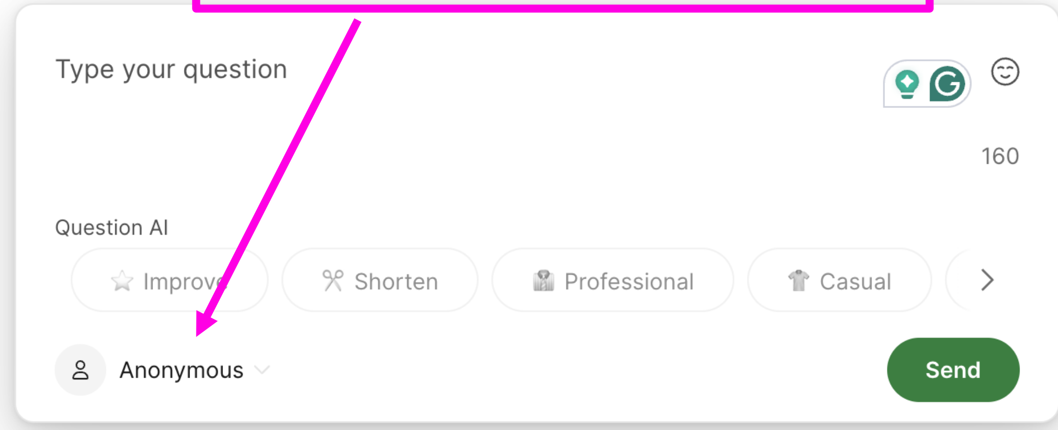

https://posit.cloud/